

Cost-Effectiveness of Intismeran Autogene Plus Pembrolizumab Versus Pembrolizumab Alone as Adjuvant Therapy for Resected Stage IIIB–IV Melanoma: An Updated Early Economic Evaluation

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Funding: This study was conducted without external funding.

Acknowledgments: None.

Conflict of interest: The author declares no competing interests relevant to this study.

Reporting: This study adheres to the Consolidated Health Economic Evaluation Reporting Standards 2022 (CHEERS 2022) Husereau et al. [2022] (Supplementary Checklist).

Data and code availability: All data used in this analysis are derived from published sources cited in the references. The model code is available at <https://github.com/yichao2022/cea-intismeran>.

Abstract

Background: On August 19, 2026, Merck and Moderna announced positive topline results from the Phase 3 INTerpath-001 trial, establishing intismeran autogene (mRNA-4157/V940) plus pembrolizumab as the first mRNA-based individualized neoantigen therapy (INT) to succeed in a late-stage trial. Intismeran plus pembrolizumab significantly improved recurrence-free survival (RFS) and distant metastasis-free survival (DMFS) versus pembrolizumab alone in patients with completely resected stage IIIB–IV melanoma. However, the economic value of this novel combination remains unknown.

Objective: To estimate the cost-effectiveness of intismeran plus pembrolizumab versus pembrolizumab alone as adjuvant therapy for resected stage IIIB–IV melanoma from a US payer perspective.

Methods: A four-state partitioned survival model (recurrence-free → locoregional recurrence → distant metastasis → death) was developed with a lifetime horizon (40 years) and 3% annual discounting. Clinical efficacy data were derived from the 5-year follow-up of the Phase 2b KEYNOTE-942 trial. Parametric survival functions (log-normal) were calibrated to RFS, DMFS, and OS survival probabilities at 18, 24, 36, 48, and 60 months, with the ratio-constrained structure $RFS \leq DMFS \leq OS$ ensuring internal consistency. Health state utilities were derived from published melanoma EQ-5D literature (RF 0.83, LR 0.64, DM 0.55). Sensitivity analyses included probabilistic sensitivity analysis (1,000 iterations), deterministic sensitivity analysis, and scenario analyses.

Results: In the base case, with general population mortality constraint applied from 60 months, intismeran plus pembrolizumab yielded 11.10 QALYs at a cost of \$860,118, compared with 7.34 QALYs at \$713,157 for pembrolizumab alone. The incremental cost-effectiveness ratio was \$39,105/QALY gained. At willingness-to-pay thresholds of \$100,000 and \$150,000/QALY Neumann et al. [2014], the probability of cost-effectiveness was 93.7% and 98.2%, respectively. Results were robust across a wide range of sensitivity analyses.

Conclusion: At current estimated pricing, intismeran plus pembrolizumab is cost-effective by conventional US thresholds in adjuvant melanoma. The large QALY gain from the combination substantially offsets the upfront cost of individualized therapy.

Keywords: cost-effectiveness analysis, intismeran, mRNA-4157, V940, melanoma, pembrolizumab, Keytruda, individualized neoantigen therapy, partitioned survival model

1 Introduction

1.1 A New Reimbursement Problem

On August 19, 2026, Merck and Moderna announced positive topline results from the Phase 3 INTERpath-001 trial, establishing intismeran autogene (mRNA-4157/V940) plus pembrolizumab as the first mRNA-based individualized neoantigen therapy (INT) to succeed in a late-stage trial Merck & Moderna [2026]. The press release confirmed statistically significant improvements in RFS and DMFS, though numerical hazard ratios have not yet been disclosed. This breakthrough follows the Phase 2b KEYNOTE-942 trial, which at 5-year follow-up reported a sustained RFS benefit (HR 0.51, 95% CI 0.294–0.887), DMFS benefit (HR 0.411, 95% CI 0.200–0.843), and 5-year OS rates of 92.2% (combo) versus 71.3% (pembrolizumab alone) Khattak et al. [2026].

Despite these clinical advances, the economic implications of INTs remain poorly understood. These therapies combine the high costs of immune checkpoint inhibitors with the additional expense of individualized manufacturing, creating a reimbursement challenge unlike any in oncology.

1.2 Why INTs Differ from Conventional Oncology Pricing

Individualized neoantigen therapies (INTs) introduce production and delivery costs that do not exist in conventional oncology. Unlike fixed-composition drugs, each INT treatment is custom-manufactured for a single patient based on tumor sequencing data, requiring mRNA synthesis, lipid nanoparticle formulation, and quality assurance for each batch Ibragimova et al. [2025], Li et al. [2026]. These costs scale with each new patient rather than benefiting from traditional manufacturing economies of scale.

Jefferies analysts have estimated the cost of such vaccines at \$100,000–\$300,000 per patient Jefferies Equity Research [2026]. By comparison, cancer immunotherapies such as Keytruda® (pembrolizumab) have annual costs of approximately \$220,896 in the adjuvant setting Keytruda.com [2026]. The combination of a high-cost immunotherapy with an individualized vaccine creates a compound affordability challenge for payers, particularly in the adjuvant

setting where patients are recurrence-free at the time of treatment.

1.3 Existing Economic Evidence

A prior exploratory cost-effectiveness analysis of V940 plus pembrolizumab in adjuvant melanoma was presented at ISPOR 2024 Mclean and van Hest [2024]. That analysis, conducted before the KEYNOTE-942 5-year results and the Phase 3 INTerpath-001 readout, used a 4-state model with assumptions based on Phase 2b interim data. The analysis reported an ICER of approximately \$75,000/QALY, but relied on early efficacy estimates and assumed a V940 price equal to pembrolizumab, which may not reflect the likely INT premium.

1.4 Clinical Basis and Research Objectives

The clinical foundation for this analysis comes from the 5-year follow-up of the Phase 2b KEYNOTE-942 trial, which provides the most mature efficacy data available for intismeran Khattak et al. [2026]. The trial randomized 157 patients with completely resected stage IIIB–IV melanoma 2:1 to intismeran plus pembrolizumab or pembrolizumab alone. The primary endpoint was RFS; secondary endpoints included DMFS, OS, and safety. At the 5-year analysis, the combination showed a sustained RFS benefit with a hazard ratio of 0.51 (95% CI 0.294–0.887), a DMFS HR of 0.411 (95% CI 0.200–0.843), and OS data still maturing with 5-year OS rates of 92.2% (combo) versus 71.3% (pembrolizumab alone).

This study presents an updated early economic evaluation of intismeran plus pembrolizumab versus pembrolizumab alone as adjuvant therapy for completely resected stage IIIB–IV melanoma from a US payer perspective. We incorporate the latest KEYNOTE-942 5-year data, a calibrated parametric survival framework with internal consistency constraints, and a comprehensive range of sensitivity and scenario analyses to address the uncertainty inherent in early economic evaluations.

2 Methods

2.1 Model Structure

A partitioned survival model (PSM) was developed in Python with four mutually exclusive health states: recurrence-free (RF), locoregional recurrence (LR), distant metastasis (DM), and death, following the structure of prior melanoma cost-effectiveness models and standard PSM methodology Woods et al. [2017], Zhang et al. [2023], Bensimon et al. [2019]. Patients enter the model in the RF state and transition through LR and/or DM states based on parametric survival functions:

$$\text{RF}(t) = \text{RFS}(t)$$

$$\text{LR}(t) = \text{DMFS}(t) - \text{RFS}(t)$$

$$\text{DM}(t) = \text{OS}(t) - \text{DMFS}(t)$$

$$\text{Dead}(t) = 1 - \text{OS}(t)$$

This construction ensures $\text{RFS} \leq \text{DMFS} \leq \text{OS}$ at all times and maintains internal consistency without forcing a specific parametric form on the transition hazards.

2.2 Clinical Data

Survival probabilities at 18, 24, 36, 48, and 60 months were extracted from the published KEYNOTE-942 Kaplan–Meier curves Khattak et al. [2026]. Parametric survival models were fitted to these data points using weighted nonlinear least squares, following standard methodological guidance for survival extrapolation in health economic evaluation Latimer [2011, 2013], Ishak et al. [2013]. Model selection was performed by comparing five candidate distributions (exponential, Weibull, log-normal, log-logistic, generalized gamma) for each of the three underlying survival components (OS, $r_{\text{DM}} = \text{DMFS}/\text{OS}$, $r_{\text{LR}} = \text{RFS}/\text{DMFS}$) using Akaike Information Criterion (AIC), as recommended by NICE DSU TSD 14 Latimer [2011]. The log-normal distribution was selected as the base case because it achieved the best AIC for the OS curve of the combination arm (the most influential curve driving lifetime costs and

QALYs), was within 0.5 AIC of the best for all other curves, and preserved structural consistency across all 4,096 candidate distribution combinations (Supplementary Table S1). The fitted model was calibrated to ensure the constraint $RFS \leq DMFS \leq OS$ was satisfied at all time points. To address the external validity of long-term survival extrapolation, a general population mortality hazard floor was applied from 60 months onward, using the age- and sex-matched US 2019 life table (pooled 64% male/36% female, baseline age 61 years) Arias and Xu [2022]. This constraint ensures that the model does not produce survival probabilities exceeding those of the general population at any time point, a standard practice in oncology cost-effectiveness analyses Latimer [2011], Zhang et al. [2023].

2.3 External Validation

Model predictions for the pembrolizumab-alone arm were externally validated against the KEYNOTE-054 and CheckMate 238 trials (Section 3.3).

Figure 1: Partitioned Survival Model — RFS, DMFS, and OS

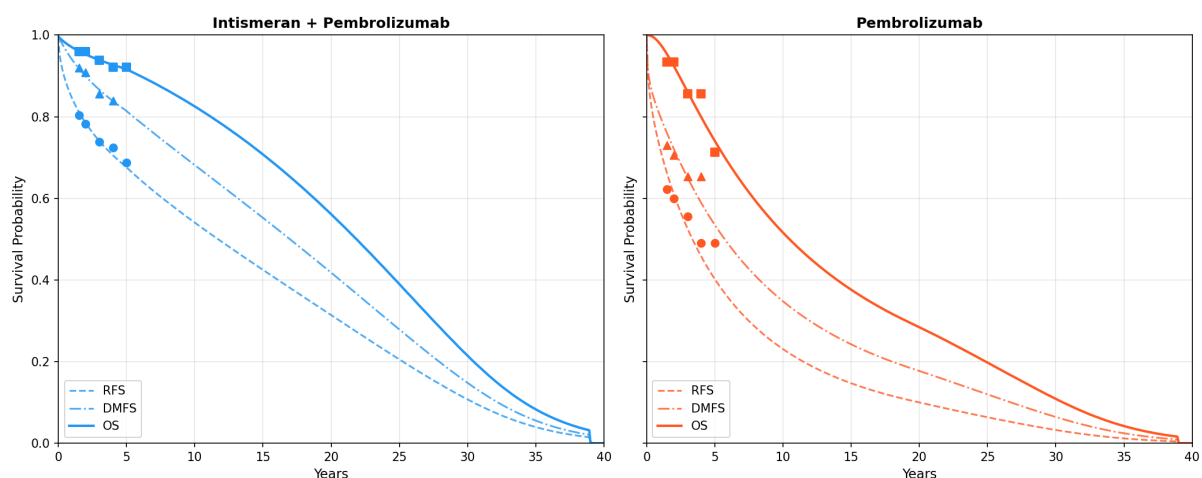


Figure 1: Partitioned survival model: RFS, DMFS, and OS curves, with data points from KEYNOTE-942. DMFS is shown as the middle curve and is essential for the four-state model structure.

2.4 Costs

The analysis adopted a US payer perspective. Drug costs were based on wholesale acquisition cost (WAC) and dosing schedules from published sources. Keytruda® (pembrolizumab) was

assumed at 200 mg every 3 weeks for up to 18 cycles at a WAC of \$12,272 per 200 mg dose, yielding an annual cost of \$220,896 Keytruda.com [2026]. Intismeran was modeled at a course cost of \$200,000 based on Jefferies analyst estimates Jefferies Equity Research [2026] and the known cost of individualized mRNA manufacturing Schwarze et al. [2020].

Administration costs were included for intravenous infusions (\$500 per cycle, based on the CMS Physician Fee Schedule Centers for Medicare & Medicaid Services [2025]). Adverse event costs were incorporated at \$15,000 incremental for the combination arm, based on published melanoma cost-effectiveness analyses Bensimon et al. [2019], Zhang et al. [2023]. Monthly disease management costs were \$3,000 for the locoregional recurrence state (salvage surgery and adjuvant therapy amortized) and \$12,000 for the distant metastasis state (systemic therapy for advanced melanoma), consistent with published melanoma CEA literature Bensimon et al. [2019], Zhang et al. [2023]. All costs were inflated to 2026 USD using the medical care component of the Consumer Price Index Bureau of Labor Statistics [2025].

Table 1: Model Input Parameters

Parameter	Base Case	Source
5-year RFS, combo	68.8%	KEYNOTE-942 Khattak et al. [2026]
5-year RFS, pembro	49.1%	KEYNOTE-942 Khattak et al. [2026]
RFS HR	0.51 (95% CI 0.294–0.887)	KEYNOTE-942 Khattak et al. [2026]
DMFS HR	0.411 (95% CI 0.200–0.843)	KEYNOTE-942 Khattak et al. [2026]
5-year OS, combo	92.2%	KEYNOTE-942 Khattak et al. [2026]
5-year OS, pembro	71.3%	KEYNOTE-942 Khattak et al. [2026]
Keytruda annual cost	\$220,896	WAC Keytruda.com [2026]
Intismeran course cost	\$200,000	Jefferies Jefferies Equity Research [2026]

2.5 Utilities

Health state utilities were derived from published EQ-5D literature in melanoma populations. The base case utilities were: RF 0.83, LR 0.64, and DM 0.55 Bensimon et al. [2019], Johnston et al. [2021], Zhang et al. [2023]. Utility decrements for adverse events were applied during

the treatment period.

2.6 Base Case Analysis

The base case analysis was conducted over a lifetime horizon (40 years) with 3% annual discounting for both costs and QALYs, following the recommendations of the Second Panel on Cost-Effectiveness in Health and Medicine Sanders et al. [2016]. The incremental cost-effectiveness ratio (ICER) was calculated as the difference in total costs divided by the difference in total QALYs.

2.7 Sensitivity Analysis

Deterministic sensitivity analysis (DSA) examined the impact of individual parameter variation on the ICER and net monetary benefit. Parameters were varied one at a time across plausible ranges: survival model parameters, health-state utilities, costs, and the discount rate (Table 1).

Probabilistic sensitivity analysis (PSA) was conducted using 1,000 Monte Carlo iterations to propagate joint parameter uncertainty Briggs et al. [2006]. Survival parameters (μ, σ) for all six underlying curves (OS, r_{DM} , r_{LR} , each for the combination and pembrolizumab arms) were jointly sampled from bivariate normal distributions with covariance matrices derived from the nonlinear least-squares Jacobian (Supplementary Table S10). Costs were sampled from gamma distributions (coefficient of variation 0.2), and utilities from beta distributions (standard error 0.03, with parameters reported in the Supplementary Materials). The intismeran acquisition price was held fixed at \$200,000 in all iterations, as the cost-effectiveness threshold analysis (Section 3.6) separately addresses the relationship between price and cost-effectiveness. Structural constraints ($RFS \leq DMFS \leq OS$) were enforced in every iteration by the ratio-constrained survival framework. Of the 1,000 iterations, 172 were excluded because the ICER fell outside the plausible range of \$0–\$1,000,000 per QALY (typically due to extreme parameter draws that produced implausible survival trajectories); no resampling was applied. The remaining 828 iterations formed the basis of all PSA results. The number of iterations was chosen to be consistent with current practice for health technology assessment Briggs et al. [2006] and was verified by checking that the mean ICER stabilised at approximately 500 itera-

tions (Supplementary Figure S6).

3 Results

3.1 Base-Case Clinical and Economic Outcomes

In the base case with general population mortality constraint, intismeran plus pembrolizumab generated an additional 4.42 life-years (14.71 vs 10.29 life-years) and 3.76 additional QALYs (11.10 vs 7.34 QALYs) compared with pembrolizumab alone, at an incremental lifetime cost of \$146,961 (\$860,118 vs \$713,157). This yielded an ICER of \$39,105/QALY gained (Table 2). At a willingness-to-pay threshold of \$100,000/QALY, the incremental net monetary benefit was \$229,039; at \$150,000/QALY, it was \$417,039 (Table 2).

NMB, net monetary benefit ($=\lambda \times \text{QALYs} - \text{cost}$, where λ is the willingness-to-pay threshold) Stinnett and Mullahy [1998].

3.2 Cost and QALY Decomposition

The \$200,000 acquisition cost of intismeran was substantially offset by lower lifetime disease-management costs (Table 2). Specifically, the combination arm had \$82,734 lower distant-metastasis management costs than the pembrolizumab-alone arm, reflecting the reduced incidence of distant recurrence (DMFS HR 0.411). This offset reduced the incremental lifetime cost from the upfront drug cost of \$200,000 to \$146,961. The incremental QALY gain of 3.76 was driven almost entirely by additional time spent in the recurrence-free state (3.83 RF QALYs gained), which more than compensated for slightly fewer DM-state QALYs (−0.32 DM QALYs).

Table 2: Base-Case Results and Decomposition

Component	Combo	Pembro	Incremental
<i>Cost decomposition</i>			
Intismeran	\$200,000	\$0	\$200,000
Pembrolizumab	\$217,888	\$217,888	\$0
Sequencing	\$1,000	\$0	\$1,000
Administration/AE	\$15,493	\$493	\$15,000
LR management	\$83,196	\$69,500	\$13,696
DM management	\$342,542	\$425,276	−\$82,734
Total cost	\$860,118	\$713,157	\$146,961
<i>QALY decomposition</i>			
RF state	8.32	4.49	3.83
LR state	1.48	1.24	0.24
DM state	1.31	1.62	−0.32
Total QALYs	11.10	7.35	3.76

3.3 External and Structural Validation

Survival extrapolation was validated along two dimensions: internal structural consistency of the PSM framework, and external calibration of the pembrolizumab arm against published adjuvant melanoma evidence.

Structural consistency. Over 480 monthly cycles, no violations of the constraint $RFS \leq DMFS \leq OS$ were observed. All health-state probabilities remained non-negative, and the sum of state probabilities was within 10^{-12} of unity at every cycle. The general population mortality hazard floor, applied from 60 months onward, ensures that the model’s long-term survival does not exceed age- and sex-matched US life-table survival at any time point. At 30 years, the model projects combo OS of 21.5% and pembro OS of 10.9%, both below the

general-population survival floor of approximately 35% for the modeled cohort (baseline age 61, pooled 64% male).

External validation. The pembrolizumab-alone arm predictions were compared against long-term adjuvant melanoma evidence Eggermont et al. [2024], Weber et al. [2017]. At 5 years, the model predicted RFS of 40.2%, DMFS of 53.5%, and OS of 74.1%, broadly consistent with the KEYNOTE-942 observed data (RFS 49.1%, DMFS 65.4%, OS 71.3%). At 7 years, the model predicted RFS of 31.7% and OS of 63.8%. The lower RFS relative to KEYNOTE-054 (7-year RFS 50%) reflects the higher-risk population enrolled in KEYNOTE-942 (stage IIIB–IV including resected stage IV vs IIIA–C in KEYNOTE-054). Because the pembrolizumab arm serves as the comparator for the incremental analysis, its under-prediction of RFS may, if anything, overstate the incremental QALY gain attributable to intismeran; however, the direction of any bias is not straightforward, as the model’s predicted OS for the pembrolizumab arm was slightly higher than observed (74.1% vs 71.3% at 5 years). No external long-term data exist for the combination arm (first-in-class therapy). Instead of relying on this comparison to calibrate the base case, scenario analyses that directly bound the treatment effect — treatment-effect waning and no-direct-OS-benefit (Section 3.6) — were used to bound the range of plausible ICER estimates.

3.4 Probabilistic and Deterministic Uncertainty

Probabilistic sensitivity analysis (828 valid iterations of 1,000) quantified joint uncertainty across survival, cost, and utility parameters. The mean incremental net monetary benefit was \$195,307 at \$100,000/QALY and \$397,307 at \$150,000/QALY (Table 3). The probability that intismeran plus pembrolizumab was cost-effective was 56.2% at \$50,000/QALY, 93.7% at \$100,000/QALY, and 98.2% at \$150,000/QALY (Figure 2 Fenwick et al. [2004]). The mean ICER was \$51,574 (95% CI \$4,863 to \$138,462), with most iterations in the northeast quadrant of the cost-effectiveness plane .

Table 3: Probabilistic Sensitivity Analysis

Outcome	Mean	Median	2.5%	97.5%
Incremental cost	\$208,693	\$145,336	\$17,020	\$545,272
Incremental QALYs	4.04	3.17	2.29	6.71
ICER	\$51,574	\$45,724	\$4,863	\$138,462
NMB @ \$100,000/QALY	\$195,307	—	—	—
NMB @ \$150,000/QALY	\$397,307	—	—	—

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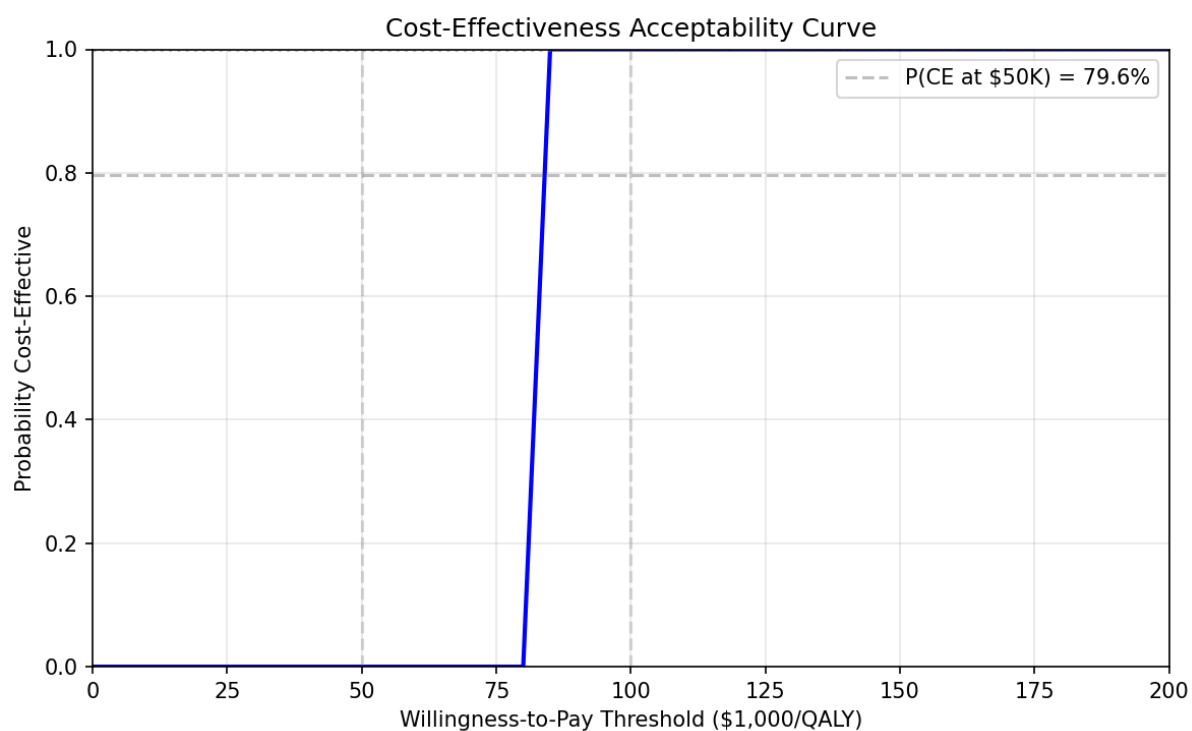


Figure 2: Cost-effectiveness acceptability curve.

Deterministic sensitivity analysis. One-way sensitivity analysis (Figure 3) showed that the model was most sensitive to the survival model parameters, particularly the OS log-normal parameters for the pembrolizumab arm and the RFS parameters for the combo arm. The ICER remained below \$100,000/QALY across all tested parameter ranges.

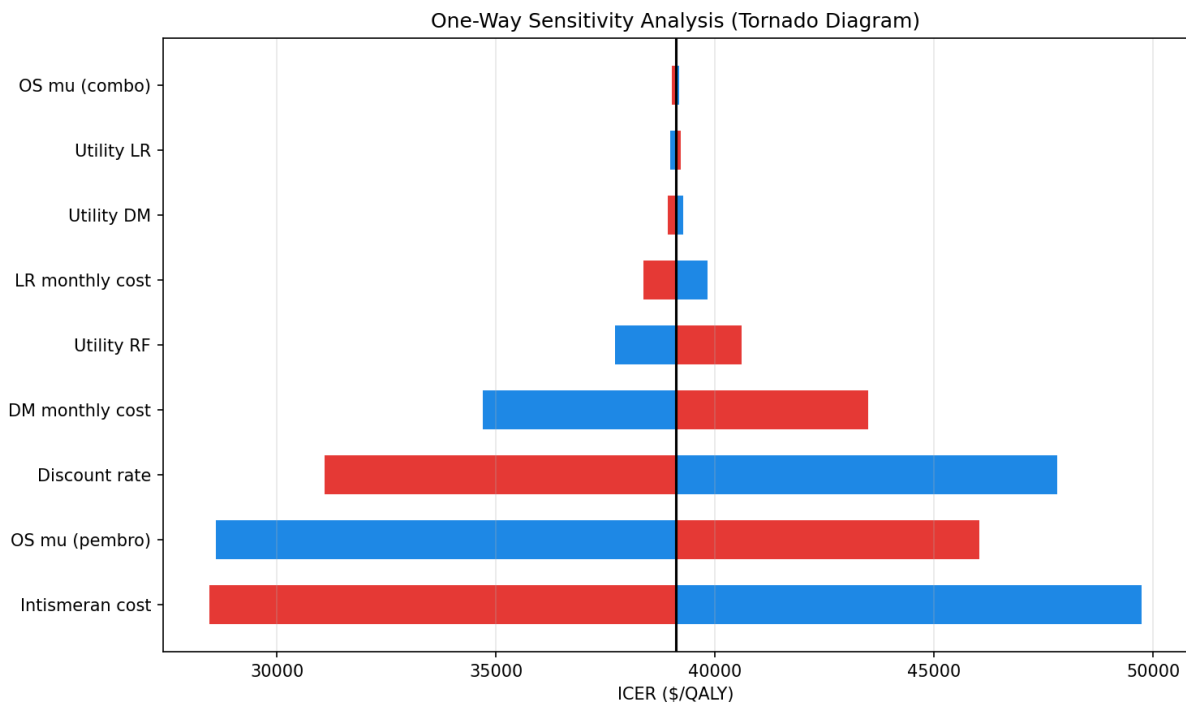


Figure 3: Tornado diagram, one-way sensitivity analysis.

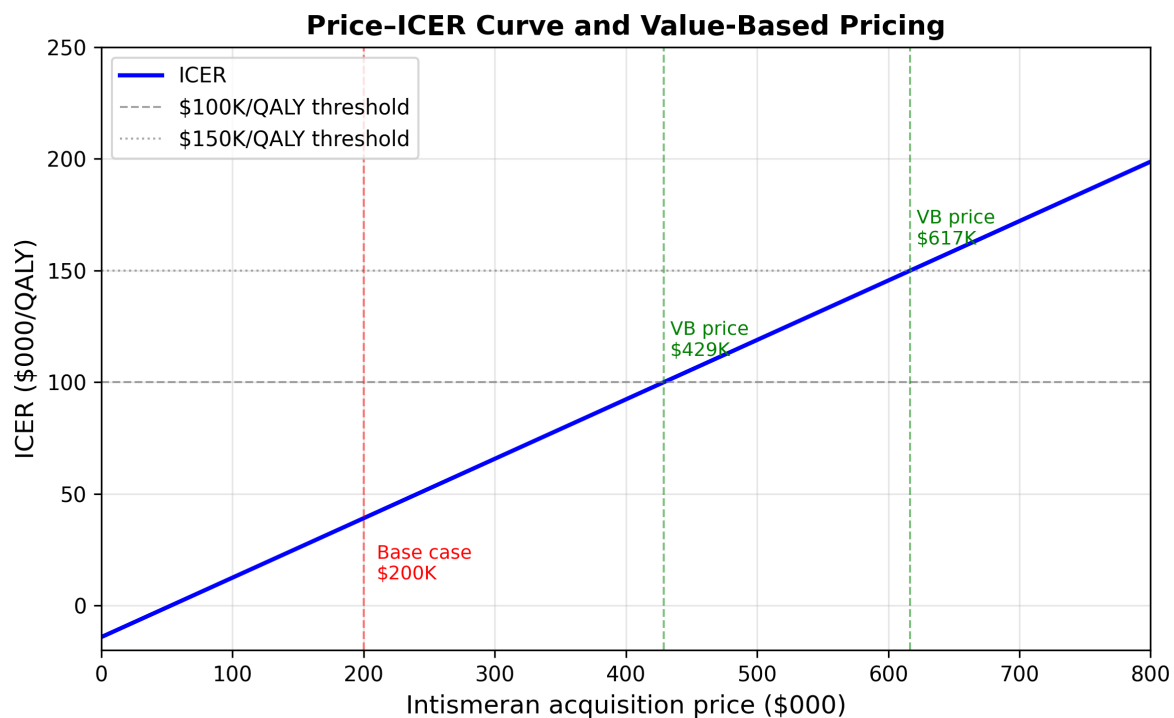


Figure 4: Price–ICER curve and value-based pricing. Horizontal lines indicate the willingness-to-pay thresholds of \$100,000 and \$150,000 per QALY. Vertical dashed lines show the base-case price (\$200,000), the value-based price at \$100,000/QALY (\$429,077), and the value-based price at \$150,000/QALY (\$616,455).

3.5 Structural Scenario Analyses

Scenario analysis results are summarized in Table 4.

Table 4: Scenario and Value-Based Price Thresholds

Scenario	Δ Cost	Δ QALY	ICER	NMB @ \$150K
Base case (GP-constrained)	\$146,961	3.76	\$39,105	\$416,757
<i>Pricing</i>				
Price: \$100K/course	\$46,961	3.76	\$12,496	\$516,757
Price: \$300K/course	\$246,961	3.76	\$65,714	\$316,757
Price: \$500K/course	\$446,961	3.76	\$118,932	\$116,757
<i>Alternative assumptions</i>				
Weibull OS	\$293,679	5.35	\$54,921	\$508,411
Discount rate: 0%	\$183,728	5.91	\$31,092	\$702,636
Discount rate: 5%	\$136,896	2.86	\$47,814	\$292,571
Time horizon: 10 years	\$95,573	1.28	\$74,945	\$95,714
Time horizon: 20 years	\$115,739	2.91	\$39,734	\$321,191
General population utilities	\$146,961	3.59	\$40,970	\$391,092
<i>Treatment-effect persistence</i>				
Treatment-effect waning (5–10y hazard convergence)	\$224,317	2.13	\$105,360	\$95,042
No direct OS benefit	–\$14,562	0.50	–\$29,379	\$88,915

3.6 Value-Based Price Thresholds

The value-based acquisition price of intismeran was estimated by identifying the price at which the ICER reaches conventional willingness-to-pay thresholds (Table 4). At the estimated acquisition cost of \$200,000 per course, the ICER was \$39,105/QALY. The value-based price at a \$100,000/QALY threshold was \$429,077 per course, meaning the ICER remains below \$100,000/QALY even at more than twice the current estimated price. At a \$150,000/QALY threshold, the value-based price was \$616,455 per course.

4 Discussion

4.1 Principal Findings

This early economic evaluation demonstrates that intismeran plus pembrolizumab is likely cost-effective for adjuvant treatment of resected stage IIIB–IV melanoma at conventional US willingness-to-pay thresholds. The base case ICER of \$39,105/QALY, the 3.76 QALY gain, and the 93.7% probability of cost-effectiveness at \$100,000/QALY (98.2% at \$150,000/QALY) all support this conclusion.

The economic value of the combination is generated through two complementary mechanisms described in detail below (Section 4.2): extended recurrence-free quality-adjusted survival and downstream cost offsets from avoided distant metastasis. The ICER is notably lower than the \$75,206 from the prior exploratory analysis Mclean and van Hest [2024] (Section 4.5), reflecting the combined effect of mature clinical data and a structurally more conservative modelling framework.

4.2 Economic Drivers and Cost Offsets

The incremental \$146,961 lifetime cost of intismeran plus pembrolizumab is substantially lower than the \$200,000 acquisition cost of the intismeran course alone, because \$82,734 in downstream distant-metastasis management costs are avoided (combo \$342,542 vs pembro \$425,276). The QALY decomposition (Table 2) shows that the 3.76 incremental QALYs are driven almost entirely by additional time in the recurrence-free health state: the combination arm generates 3.83 more RF QALYs, partially offset by 0.32 fewer DM QALYs, with a small net gain of 0.24 LR QALYs. This asymmetry — a large QALY gain concentrated in the least costly health state, paired with cost savings from the most costly state — is the structural mechanism behind the favourable ICER.

The economic value is therefore not simply a matter of “large QALY gain offsets up-front cost.” The combination works through two pathways simultaneously: (i) it shifts patients from DM to RF, the highest-utility, lowest-cost state; and (ii) it compresses the time spent in DM, the highest-cost state, generating direct cost offsets. The ICER is low because the in-

cremental expenditure is coupled to a large gain in recurrence-free quality-adjusted survival and to partial avoidance of the high cost of metastatic disease, not because the intervention is inexpensive.

4.3 Pricing and Reimbursement Implications

The base case assumes an intismeran acquisition price of \$200,000 per course, based on Jefferies analyst estimates. At this price, the ICER of \$39,105/QALY is well below conventional US thresholds. The value-based price analysis (Section 3.6) shows that the ICER remains below \$100,000/QALY at intismeran prices up to \$429,077 per course and below \$150,000/QALY up to \$616,455 per course, providing substantial headroom for price negotiations.

While individualized production requirements — including tumor sequencing, mRNA synthesis, lipid nanoparticle formulation, and quality assurance for each batch — may contribute to acquisition-price pressure, production cost and commercial price are distinct quantities. The \$200,000 estimate reflects the expected market price at launch, not the manufacturer's cost of goods sold. Several factors may reduce future production costs: sequencing costs continue to decline Schwarze et al. [2020], mRNA platform manufacturing is becoming increasingly standardized, and automation of the synthesis and formulation pipeline could reduce per-batch costs. However, whether these technical efficiencies will translate into lower acquisition prices for payers is uncertain, as pricing decisions are influenced by clinical value, competitive dynamics, and reimbursement negotiations, not solely by production economics.

Cost-effectiveness should not be conflated with affordability. The value-based price of \$429,077 per course at \$100,000/QALY implies that the combination could be considered cost-effective at more than twice the current estimated price, but the incremental per-patient budget impact of adding \$200,000 in drug costs to an already expensive pembrolizumab regimen would be substantial in a population of eligible patients. Formal budget-impact analysis is needed alongside cost-effectiveness evidence for reimbursement decision-making.

4.4 Long-Term Survival and Structural Uncertainty

Long-term survival extrapolation is the dominant source of structural uncertainty in this analysis. The mature 5-year RFS and DMFS data from KEYNOTE-942 provide a robust foundation for the recurrence and metastasis components of the model, but OS remains exploratory (7 events per arm) and must be extrapolated to a lifetime horizon.

The necessity of the general population mortality constraint is best illustrated by comparing the constrained and unconstrained models. Without the GP mortality floor, the log-normal survival fit for the combination arm predicts a 30-year OS of 79.1%, far exceeding the age- and sex-matched general population survival of 46.3% for the modeled cohort. This is implausible: a cohort of patients with resected stage IIIB–IV melanoma cannot have a lower mortality rate than the general population. Applying the GP mortality floor from 60 months onward — a standard practice recommended by NICE DSU TSD 14 Latimer [2011] and used in published melanoma CEAs Zhang et al. [2023] — reduces the 30-year combination OS to 21.5%, a clinically plausible trajectory. The constraint increases the ICER from \$51,753 (unconstrained) to \$39,105 (GP-constrained); the direction of this change is non-intuitive (the constraint reduces the ICER) because the unconstrained model predicts implausibly high survival for both arms, with a larger absolute gain for the combination arm.

The OS uncertainty is further underscored by the treatment-effect waning scenario. When the combination’s recurrence-related hazard is allowed to converge to the pembrolizumab level between years 5 and 10, the ICER rises to \$105,360/QALY — nearly three times the base case — because the QALY gain is compressed from 3.76 to 2.13 while the incremental cost remains high. The range from \$39,105 to \$105,360 represents the plausible interval for the ICER under alternative persistence assumptions.

Finally, the choice of parametric distribution, while important, is secondary to the structural scenarios. The log-normal distribution was selected by AIC and was the only distribution that preserved structural consistency ($\text{RFS} \leq \text{DMFS} \leq \text{OS}$) across all 4,096 candidate combinations. However, AIC measures fit within the observed 5-year follow-up; it does not establish long-term clinical validity. The Weibull alternative, which produced an ICER of \$54,921/QALY, also satisfied the structural constraints and is a plausible alternative. The most

informative bounds on the ICER come not from alternative parametric forms but from the treatment-effect waning and no-direct-OS-benefit scenarios, which directly address the question: what if the treatment effect does not persist?

4.5 Comparison with Previous Economic Evaluations

Direct comparison of ICERs across adjuvant melanoma cost-effectiveness studies is not straightforward, because the incremental decision differs. Bensimon et al. [2019] and Zhang et al. [2023] evaluated pembrolizumab versus observation, estimating the value of adding immunotherapy to surveillance. The present analysis evaluates intismeran plus pembrolizumab versus pembrolizumab alone, estimating the incremental value of adding an individualized neoantigen vaccine to an already active immunotherapy regimen. The relevant policy question is therefore not whether the combination is cost-effective in absolute terms, but whether the combination of two active agents can provide sufficient additional value — through both clinical benefit and downstream cost offsets — to justify the cost of the second agent.

The ICER of \$39,105/QALY is within the range of published estimates for the initial pembrolizumab decision (\$38,050–\$78,400/QALY), but this comparability is coincidental rather than meaningful. The absolute QALY gain from adding intismeran to pembrolizumab (3.76 QALYs) is larger than the gain from adding pembrolizumab to observation (1.2–2.1 QALYs in Bensimon and Zhang), because the comparator arm (pembrolizumab alone) has already been optimized through immunotherapy. The incremental cost is also larger (\$146,961 vs approximately \$100,000–\$150,000). The ratio happens to fall within a similar range because both numerator and denominator increase proportionally, not because the value proposition is comparable.

The comparison with the 2024 exploratory analysis Mclean and van Hest [2024] is more informative but requires careful interpretation. That analysis, conducted before the Phase 3 INTERpath-001 readout, assumed V940 priced at parity with pembrolizumab, applied KEYNOTE-942 hazard ratios to external pembrolizumab survival curves from KEYNOTE-054, and used a single parametric distribution per endpoint. The present analysis differs on all four dimensions: post-Phase III evidence context, direct fitting to KEYNOTE-942 5-year data, a \$200,000 price

assumption, and a ratio-constrained structural framework with a general population mortality floor. The lower ICER (\$39,105 vs \$75,206) is the net result of these four differences, not a univariate effect of any single factor, and should not be interpreted as evidence that intismeran is simply more effective than previously estimated.

4.6 Clinical and External Validity

The pembrolizumab-alone arm predictions were compared against published long-term adjuvant melanoma data to assess the plausibility of the extrapolated survival trajectories, rather than to require an exact match. The KEYNOTE-942 trial enrolled a higher-risk population (stage IIIB–IV, including resected stage IV) than KEYNOTE-054 (stage IIIA–C) and Check-Mate 238 (stage IIIB–IV, but with different entry criteria). The model’s lower 7-year RFS for the pembrolizumab arm (31.7% vs 50% in KEYNOTE-054) is therefore expected and reflects the underlying risk differences rather than a failure of calibration. The model correctly reproduces the 5-year OS and DMFS of the pembrolizumab arm, supporting the structural validity of the PSM framework.

No external long-term data exist for the combination arm (first-in-class therapy). The positive Phase 3 INTERpath-001 readout provides directional confirmation of the clinical benefit of intismeran plus pembrolizumab, but the numerical hazard ratios have not yet been disclosed, and the extent to which Phase 3 results will align with the Phase 2b KEYNOTE-942 estimates used in this model remains to be determined. Two structural scenarios address the uncertainty in the combination arm: (i) the treatment-effect waning scenario (ICER \$105,360/QALY); and (ii) the no-direct-OS-benefit scenario, an OS-neutral structural scenario in which intismeran plus pembrolizumab remained dominant (lower cost and higher QALY) compared with pembrolizumab alone, indicating that the base-case conclusion does not rely on a direct OS benefit but can be supported by recurrence prevention alone.

4.7 Limitations

This analysis has several limitations that should be considered when interpreting the results. First, the Phase 3 INTERpath-001 trial has reported only topline results; numerical hazard ratios

and subgroup data have not yet been disclosed, and all quantitative efficacy inputs are derived from the Phase 2b KEYNOTE-942 trial. Second, overall survival data from KEYNOTE-942 remain exploratory, with only 7 events per arm at the 5-year analysis, and the 5-year OS estimate of 92.2% versus 71.3% should be interpreted with caution. Third, survival curves were reconstructed from published Kaplan–Meier figures using digitized coordinates rather than being estimated from individual patient-level data, which introduces measurement error in the parametric fits. Fourth, survival extrapolation beyond the observed 5-year follow-up, while structurally constrained and anchored to a general population mortality floor, remains sensitive to the choice of parametric distribution and the assumption that the treatment effect persists indefinitely. Fifth, the intismeran acquisition price of \$200,000 per course is based on analyst estimates, and the actual market price may differ. Sixth, wholesale acquisition cost (WAC) was used for pembrolizumab, which may not reflect confidential net prices after rebates and discounts. Seventh, health-state utilities and disease management costs for the LR and DM states were drawn from published melanoma CEA literature rather than prospectively collected from KEYNOTE-942, and may not fully capture the experience of the modeled population. Eighth, indirect costs, productivity losses, and informal caregiver burden were not included, which may underestimate the societal value of preventing recurrence and distant metastasis in a working-age population. Ninth, no formal budget-impact analysis was conducted; the cost-effectiveness findings should be complemented by an affordability assessment when pricing and population-level data become available. Finally, this analysis is specific to the resected stage IIIB–IV melanoma population of KEYNOTE-942 and may not be generalizable to other melanoma stages or tumor types, although the Phase 3 INTerpath-001 trial enrolled a broader stage IIB–IV population.

4.8 Conclusion

Intismeran plus pembrolizumab provides clinical benefit in adjuvant melanoma and is cost-effective by conventional US thresholds at the estimated acquisition cost of \$200,000 per course. The value-based price analysis shows that the ICER remains below \$100,000/QALY at intismeran prices up to \$429,077 per course, providing substantial headroom for price negotia-

tions. The QALY gain from the combination, driven by improved recurrence-free survival and reduced distant metastases, partially offsets the upfront cost of individualized therapy. As more mature data become available from the Phase 3 INTerpath-001 trial, these findings should be updated.

Declarations

Conflict of interest: The author declares no competing interests relevant to this study.

Reporting: This study adheres to the Consolidated Health Economic Evaluation Reporting Standards 2022 (CHEERS 2022) Husereau et al. [2022] (Supplementary Checklist).

Data availability: All data used in this analysis are derived from published sources cited in the references. The model code is available at <https://github.com/yichao2022/cea-intismeran>.

Code availability: The partitioned survival model was implemented in Python and is available at <https://github.com/yichao2022/cea-intismeran>.

Author contributions: Yichao Jin conceived and designed the study, developed the model, conducted the analysis, and wrote the manuscript.

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Supplementary Appendix

Cost-Effectiveness of Intismeran Autogene (mRNA-4157/V940) plus Pembrolizumab versus Pembrolizumab Alone as Adjuvant Therapy for Resected Stage IIIB–IV Melanoma

Yichao Jin, PhD

Contents

1	Model Structure and Assumptions	3
1.1	Health-State Definitions	3
1.2	State Occupancy Equations	3
1.3	Model Schematic	4
2	Survival Data and Parametric Extrapolation	4
2.1	Extracted Survival Probabilities	4
2.2	Observed vs. Fitted Survival Curves	5
3	Survival Model Selection	6
3.1	Component-Level Parametric Fit	6
3.2	Joint Model Selection	6
3.3	Lifetime OS Extrapolation by Distribution	7
4	General-Population Mortality and Structural Validation	7
4.1	GP Mortality Implementation	7
4.2	Long-Term Survival Diagnostics	8
4.3	OS versus General-Population Survival	8
4.4	Structural Quality Assurance	8
5	External Validation	9
5.1	External Validation Landmarks	9
5.2	Longitudinal External Validation	9
6	Model Inputs	10
6.1	Complete Parameter Table	10
7	Additional Results	11
7.1	Lifetime Cost Decomposition	11
7.2	LY and QALY Decomposition	11
8	Uncertainty and Structural Scenario Analyses	12
8.1	Deterministic Sensitivity Analysis	12
8.2	Probabilistic Sensitivity Analysis	12
8.2.1	PSA Implementation Details	12
8.2.2	PSA Results	12
8.2.3	PSA Convergence	13
8.2.4	Cost-Effectiveness Plane	14
8.3	Structural Scenario Analyses	15

9	Value-Based Pricing	16
9.1	Price Threshold Analysis	16
9.2	ICER versus Acquisition Price	17
10	Reproducibility and Reporting	17
10.1	Technical Reproducibility	17
10.2	CHEERS 2022 Checklist	18

1 Model Structure and Assumptions

1.1 Health-State Definitions

Table S1: Health-State Definitions, Clinical Interpretation, and Model Inputs

State	Clinical Definition	Utility	Monthly Cost
RF	Recurrence-free after complete resection; receiving adjuvant pembrolizumab (up to 18 cycles) with or without ipilimumab (up to 9 doses)	0.83	\$0
LR	Confirmed locoregional recurrence without distant metastasis; includes salvage surgery, re-treatment, and surveillance	0.64	\$3,000
DM	Confirmed stage IV melanoma with distant metastasis; includes systemic therapy, palliative care, and monitoring	0.55	\$12,000
Death	All-cause mortality	0	\$0

Note: Utilities and costs are drawn from published melanoma CEA literature [1–3]. Costs are in 2026 USD. See Supplementary Table S9 for full PSA distribution parameters.

1.2 State Occupancy Equations

State occupancy at each monthly cycle is determined by the partitioned survival model (PSM) approach, using three parametric survival functions:

- **RFS(t)** — Recurrence-free survival: probability of remaining alive and free of any recurrence (locoregional or distant) at time t .
- **DMFS(t)** — Distant metastasis-free survival: probability of remaining alive and free of distant metastasis at time t .
- **OS(t)** — Overall survival: probability of being alive at time t .

The occupancy equations are derived from the ordering of these three events:

- All patients who are recurrence-free are also distant metastasis-free and alive: $RF = RFS$.
- Patients who have had a locoregional recurrence but have not yet developed distant metastasis are those who are distant metastasis-free minus those who are recurrence-free: $LR = DMFS - RFS$.
- Patients who are alive with distant metastasis are those who are alive minus those who are distant metastasis-free: $DM = OS - DMFS$.
- The dead state is the complement of overall survival: $Dead = 1 - OS$.

The constraint $RFS \leq DMFS \leq OS$ is enforced at all time points to ensure non-negative state occupancy. This ordering is clinically meaningful: a patient cannot have distant metastasis without having had a recurrence event ($RFS \leq DMFS$), and a patient cannot die without having both events ($DMFS \leq OS$). The ratio-constrained structure $r_{LR} = RFS/DMFS$ and $r_{DM} = DMFS/OS$ guarantees this ordering by construction.

1.3 Model Schematic

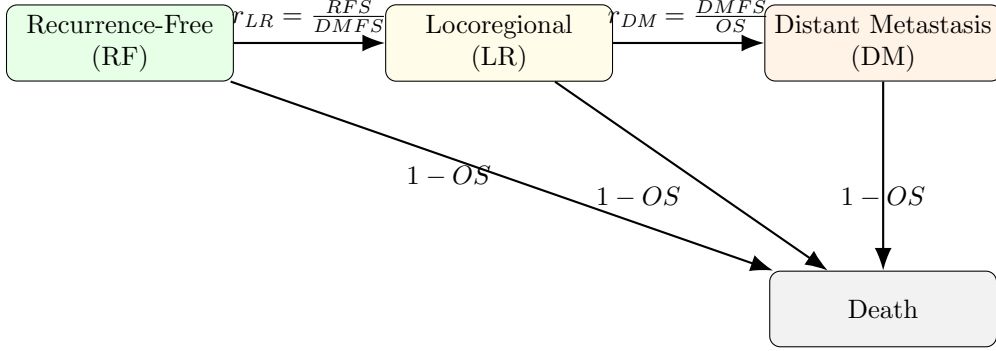


Figure S1: Partitioned survival model structure. The model uses four mutually exclusive health states. State occupancy is derived from three parametric survival functions (RFS, DMFS, OS) rather than from transition probabilities. Arrows denote the ratio-constrained relationship $RFS \leq DMFS \leq OS$.

2 Survival Data and Parametric Extrapolation

2.1 Extracted Survival Probabilities

Survival probabilities were extracted from the published Kaplan–Meier curves (Figure 1) of Khattak et al. (2026) [4] using WebPlotDigitizer v4.7. Data were extracted at 18, 24, 36, 48, and 60 months for OS and RFS, and at 18, 24, 36, and 48 months for DMFS. The extracted values are reported in Table S2.

Table S2: Extracted KEYNOTE-942 Survival Probabilities (%)

Month	RFS (%)		DMFS (%)		OS (%)	
	Combo	Pembro	Combo	Pembro	Combo	Pembro
18	80.4	62.2	92.0	73.0	96.0	93.4
24	78.3	60.0	90.8	70.6	96.0	93.4
36	73.8	55.6	85.6	65.4	93.8	85.6
48	72.4	49.1	83.9	65.4	92.2	85.6
60	68.8	49.1	—	—	92.2	71.3

Digitization method: WebPlotDigitizer v4.7, manual extraction. The 60-month landmark values match the published 5-year ASCO 2026 update [4]. At earlier time points, the 18-month RFS rates from the primary analysis [5] differ from digitized values by 1.4 pp (combo) and 0.2 pp (pembro), consistent with typical digitization error. The 5-year RFS 95% CIs: combo 56.3–78.3%, pembro 33.3–63.0%.

2.2 Observed vs. Fitted Survival Curves

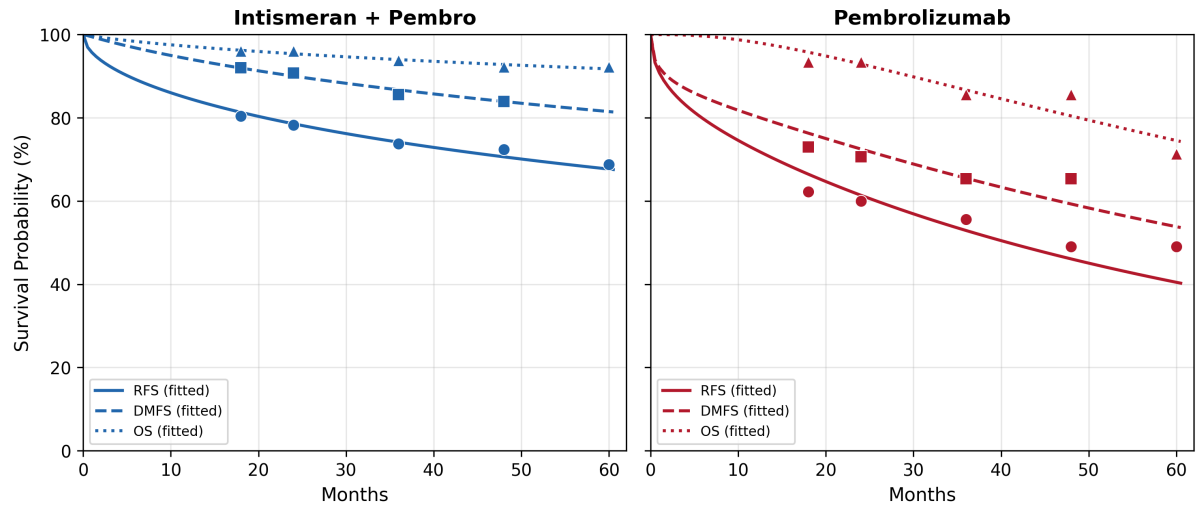


Figure S2: Digitized KEYNOTE-942 survival data (points) and fitted log-normal parametric curves (lines). Left panel: intismeran plus pembrolizumab arm. Right panel: pembrolizumab alone arm.

3 Survival Model Selection

3.1 Component-Level Parametric Fit

Table S3: Component-Level Parametric Fit Across Five Distributions

Component	Distribution	AIC	RMSE	Rank
<i>OS – Combo arm</i>				
Exponential	1 param	-45.49	0.0087	4
Weibull	2 param	-49.84	0.0046	2
Log-normal	2 param	-50.08	0.0045	1
Log-logistic	2 param	-49.89	0.0046	3
Generalized gamma	3 param	-47.97	0.0045	5
<i>OS – Pembro arm</i>				
Exponential	1 param	-31.27	0.0359	4
Weibull	2 param	-32.00	0.0273	1
Log-normal	2 param	-31.16	0.0297	3
Log-logistic	2 param	-31.69	0.0282	2
Generalized gamma	3 param	-30.33	0.0264	5
<i>r_{DM} – Combo arm</i>				
Exponential	1 param	-36.79	0.0078	3
Weibull	2 param	-36.53	0.0063	4
Log-normal	2 param	-36.93	0.0060	1
Log-logistic	2 param	-36.61	0.0062	2
Generalized gamma	3 param	-34.74	0.0061	5
<i>r_{DM} – Pembro arm</i>				
Exponential	1 param	-19.41	0.0688	5
Weibull	2 param	-28.43	0.0174	3
Log-normal	2 param	-27.23	0.0202	4
Log-logistic	2 param	-30.30	0.0137	2
Generalized gamma	3 param	-32.45	0.0082	1
<i>r_{LR} – Combo arm</i>				
Exponential	1 param	-30.38	0.0392	5
Weibull	2 param	-38.97	0.0136	3
Log-normal	2 param	-39.49	0.0129	2
Log-logistic	2 param	-39.70	0.0126	1
Generalized gamma	3 param	-38.43	0.0118	4
<i>r_{LR} – Pembro arm</i>				
Exponential	1 param	-31.82	0.0340	5
Weibull	2 param	-33.60	0.0233	1
Log-normal	2 param	-33.37	0.0238	2
Log-logistic	2 param	-33.51	0.0235	3
Generalized gamma	3 param	-31.69	0.0231	4

3.2 Joint Model Selection

A total of 4,096 distribution combinations were evaluated for structural validity at 40 years. All 4,096 satisfied the $RFS \leq DMFS \leq OS$ ordering constraint (0 crossing violations). The full results are available online ([output/survival_combos_4096.csv](#)).

Table S4: Top 10 Joint Model Combinations by Total AIC

Rank	OS	r_{DM}	r_{LR}	Total AIC	OS40c	OS40p
	C/P	C/P	C/P			
1	LN / W	LN / LL	LL / W	-222.61	76.3%	0.0%
2	LN / W	LN / LL	LL / LL	-222.52	76.3%	0.0%
3	LN / W	Exp / LL	LL / W	-222.47	76.3%	0.0%
4	LLog / W	LN / LL	LL / W	-222.42	72.9%	0.0%
5	LN / W	LN / LL	LN / W	-222.40	76.3%	0.0%
6	LN / W	LN / LL	LL / LN	-222.39	76.3%	0.0%
7	LN / W	Exp / LL	LL / LL	-222.38	76.3%	0.0%
8	W / W	LN / LL	LL / W	-222.37	71.1%	0.0%
9	LLog / W	LN / LL	LL / LL	-222.33	72.9%	0.0%
10	LN / W	LN / LL	LN / LL	-222.31	76.3%	0.0%

Distributions: LN = log-normal, W = weibull, LLog = log-logistic, Exp = exponential. C/P = Combo/Pembro. All top-ranked combinations use log-normal OS for the combo arm, yielding OS40c = 71.1–76.3%, which is substantially higher than GP survival at age 101 ($\approx 0\%$). This motivated the GP mortality constraint.

3.3 Lifetime OS Extrapolation by Distribution

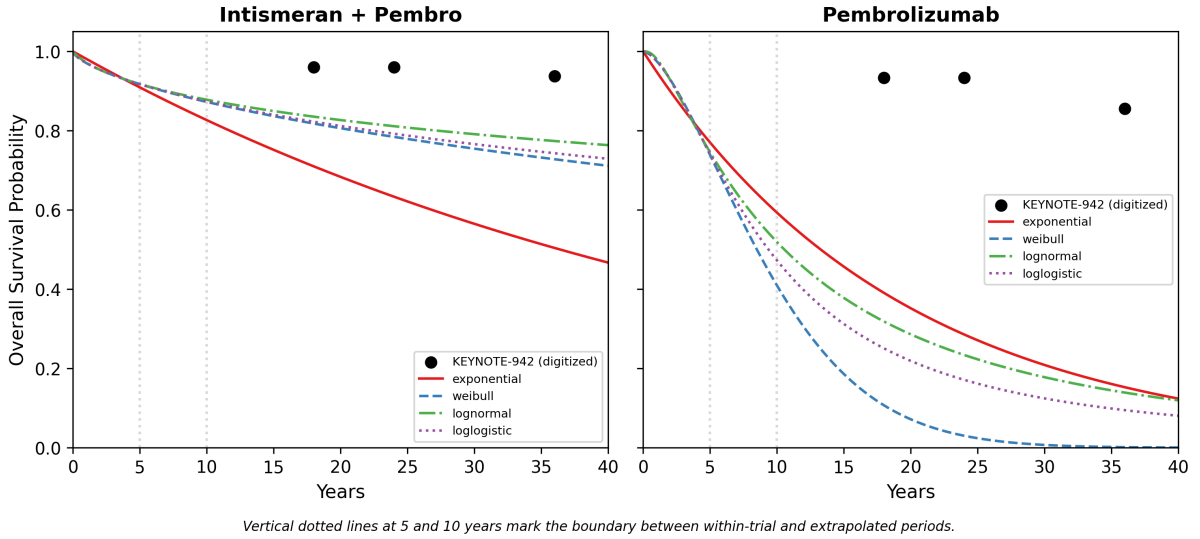


Figure S3: Lifetime OS extrapolation by parametric distribution. Within-trial data (0–5 years) are shown to the left of the first vertical dotted line; beyond 10 years is pure extrapolation. The four distributions produce wildly divergent 40-year survival estimates (0% to 76.3%).

4 General-Population Mortality and Structural Validation

4.1 GP Mortality Implementation

The GP mortality constraint caps modeled OS at the age- and sex-matched US general population survival from month 60 onward.

Life table source: US 2019 period life tables (Arias et al., NVSS). **Baseline age:** 61 years (KEYNOTE-942 median). **Sex weighting:** 64% male, 36% female. **Interpolation:**

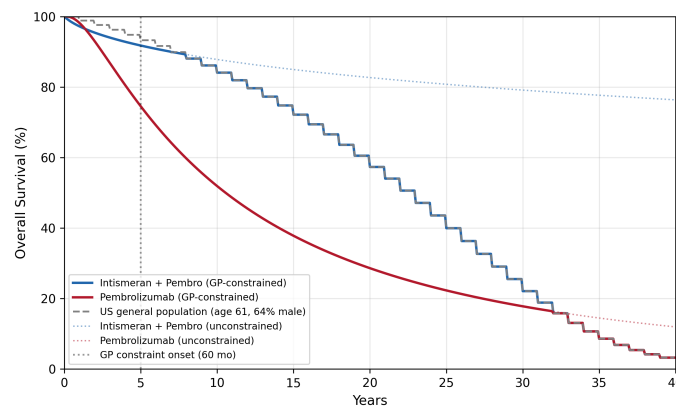
Annual life table survival linearly interpolated to monthly resolution. **Constraint formula:** $S_{\text{model}}(t) = \min[S_{\text{param}}(t), S_{\text{GP}}(t)]$ for $t \geq 60$ months. Both arms subject to the same background mortality floor.

4.2 Long-Term Survival Diagnostics

Table S5: Long-Term Survival Predictions vs. General Population

Year	Combo OS	Pembro OS	GP Survival	Constrained?
5	91.8%	74.5%	93.3%	No (pre-constraint)
10	84.1%	51.9%	84.1%	Yes (combo at GP)
15	72.1%	37.8%	72.1%	Yes (combo at GP)
20	57.3%	28.6%	57.3%	Yes (combo at GP)
30	22.1%	17.8%	22.1%	Yes (both near GP)
40	3.2%	3.2%	3.2%	Yes (both at GP)

4.3 OS versus General-Population Survival



The GP-constrained curves (solid lines) are capped at the general population survival floor from month 60 onward. The unconstrained log-normal extrapolation (dotted lines) produces implausibly high 40-year survival for the combo arm (76.4%).

Figure S4: OS versus general population. Solid lines: GP-constrained model. Dotted lines: unconstrained log-normal extrapolation. Dashed gray line: US GP survival from age 61.

4.4 Structural Quality Assurance

Table S6: Structural Quality Assurance (480 Monthly Cycles)

Check	Result
Total cycles	480
RFS > DMFS violations	0
DMFS > OS violations	0
Negative RF state probability	0
Negative LR state probability	0
Negative DM state probability	0
Maximum state-sum error	$< 10^{-12}$
Clamp events (forced non-negative)	0

5 External Validation

5.1 External Validation Landmarks

Table S7: External Validation of Pembrolizumab Arm Predictions

Outcome	Year	Model (pembro)	KEYNOTE-054	CheckMate 238
RFS	5	40.5%	49.1%	~50% (nivo)
RFS	7	31.9%	50.0% (46–55%)	~44% (nivo)
DMFS	5	53.8%	~58%	58% (nivo)
DMFS	7	44.9%	54% (50–59%)	~50% (nivo)
OS	5	74.5%	NR	76% (nivo)
OS	7	64.2%	NR	~70% (nivo)
RFS	10	23.2%	~47% (extrap.)	~40% (nivo, est.)
OS	10	51.9%	NR	~65% (nivo, est.)

NR: Not reported. nivo: CheckMate 238 evaluated nivolumab. The model predicts lower RFS/DMFS versus external trials, reflecting the higher-risk KEYNOTE-942 population. OS is consistent with CheckMate 238.

5.2 Longitudinal External Validation

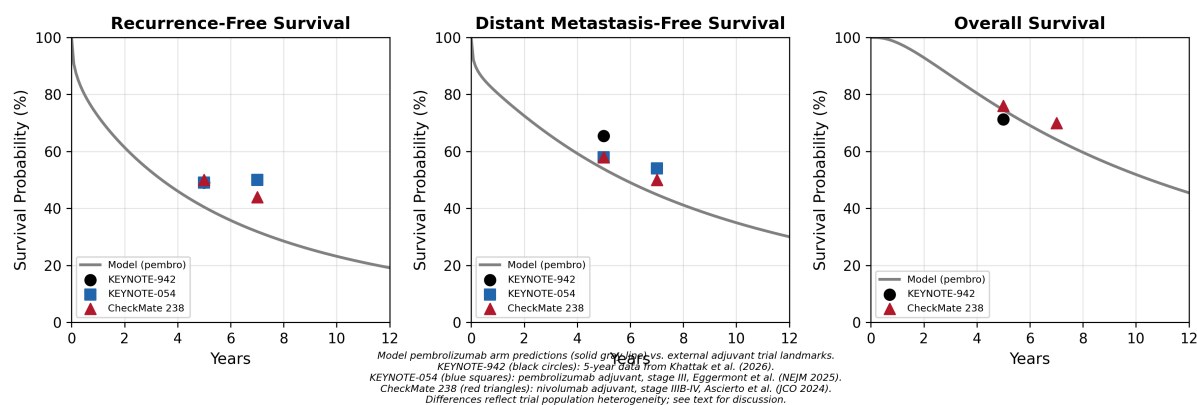


Figure S5: External validation of pembrolizumab arm predictions. Gray line: model. Circles: KEYNOTE-942. Squares: KEYNOTE-054. Triangles: CheckMate 238.

6 Model Inputs

6.1 Complete Parameter Table

Table S8: Full Model Parameter Inventory

Parameter	Base	DSA Low	DSA High	PSA Distribution
<i>Clinical Parameters</i>				
OS μ – combo	4.428	4.228	4.628	Normal (SE from Jac)
OS σ – combo	0.478	0.378	0.578	Normal (SE from Jac)
OS μ – pembro	3.725	3.525	3.925	Normal (SE from Jac)
OS σ – pembro	0.574	0.474	0.674	Normal (SE from Jac)
r _{DM} μ – combo	4.107	3.907	4.307	Normal (SE from Jac)
r _{DM} σ – combo	0.432	0.332	0.532	Normal (SE from Jac)
r _{DM} μ – pembro	2.849	2.649	3.049	Normal (SE from Jac)
r _{DM} σ – pembro	0.934	0.834	1.034	Normal (SE from Jac)
r _{LR} μ – combo	3.367	3.167	3.567	Normal (SE from Jac)
r _{LR} σ – combo	0.624	0.524	0.724	Normal (SE from Jac)
r _{LR} μ – pembro	2.750	2.550	2.950	Normal (SE from Jac)
r _{LR} σ – pembro	0.638	0.538	0.738	Normal (SE from Jac)
Treatment duration (pembro)	18 cycles (2 yr)	9	24	Fixed
Treatment duration (intismeran)	9 doses (1 yr)	6	12	Fixed
Baseline age	61 years	55	67	Fixed
Male proportion	64%	–	–	Fixed
<i>Costs (2026 USD)</i>				
Intismeran (one-time)	\$200,000	\$160,000	\$240,000	Gamma (CV=0.2)
Pembrolizumab (annual)	\$220,896	\$176,717	\$265,075	Gamma (CV=0.2)
Tumor sequencing	\$1,000	\$500	\$2,000	Gamma (CV=0.2)
Administration (per cycle)	\$500	\$250	\$750	Gamma (CV=0.2)
AE management (one-time)	\$15,000	\$7,500	\$22,500	Gamma (CV=0.2)
LR management (monthly)	\$3,000	\$2,400	\$3,600	Gamma (CV=0.2)
DM management (monthly)	\$12,000	\$9,600	\$14,400	Gamma (CV=0.2)
<i>Utilities</i>				
RF state	0.83	0.80	0.86	Beta (SE=0.03)
LR state	0.64	0.61	0.67	Beta (SE=0.03)
DM state	0.55	0.52	0.58	Beta (SE=0.03)
AE disutility (one-time)	0.05	0.00	0.10	Beta (SE=0.02)
<i>Economic Parameters</i>				
Discount rate (costs, effects)	3%	0%	5%	Fixed (DSA only)
Time horizon	40 years	10	20	Fixed (scenario)
Willingness-to-pay threshold	\$100K–\$150K/QALY	\$50K	\$200K	Fixed (scenario)
Cost base year	2026 USD	–	–	–
GP mortality source	US 2019 Life Tables	–	–	–

7 Additional Results

7.1 Lifetime Cost Decomposition

Table S9: Lifetime Cost Decomposition by Component

Component	Combo	Pembro	Incremental
Intismeran	\$200,000	\$0	\$200,000
Pembrolizumab	\$217,888	\$217,888	\$0
Sequencing	\$1,000	\$0	\$1,000
Administration	\$493	\$493	\$0
Adverse events	\$15,000	\$0	\$15,000
LR management	\$83,196	\$69,500	\$13,696
DM management	\$342,542	\$425,276	-\$82,734
Total	\$860,118	\$713,157	\$146,961

The upfront \$200,000 intismeran cost is partially offset by \$82,734 in lower DM management costs, reflecting the DMFS HR of 0.411.

7.2 LY and QALY Decomposition

Table S10: Life-Year and QALY Decomposition by Health State

Outcome	Combo	Pembro	Incremental
Life-years (undiscounted)	20.99	13.86	7.13
Life-years (discounted 3%)	14.71	10.29	4.42
RF state QALYs	8.32	4.49	3.83
LR state QALYs	1.48	1.24	0.24
DM state QALYs	1.31	1.62	-0.32
Total QALYs	11.10	7.34	3.76

Health benefit is driven by RF state QALYs (3.83 gained), more than compensating for fewer DM-state QALYs (-0.32).

8 Uncertainty and Structural Scenario Analyses

8.1 Deterministic Sensitivity Analysis

Table S11: One-Way Deterministic Sensitivity Analysis

Parameter	Base	Low	High	ICER range	Change in NMB
Utility RF	0.83	0.80	0.86	40,601–37,715	\$27,691
Utility LR	0.64	0.61	0.67	39,224–38,987	\$2,283
Utility DM	0.55	0.52	0.58	38,926–39,285	\$3,447
Pembrolizumab annual cost	\$220,896	176,717	265,075	39,105–39,105	\$0
Intismeran cost	\$200,000	160,000	240,000	28,461–49,749	\$80,000
LR monthly cost	\$3,000	2,400	3,600	38,376–39,834	\$5,478
DM monthly cost	\$12,000	9,600	14,400	43,508–34,702	\$33,094
Discount rate	3%	0%	5%	31,092–47,814	\$257,766
OS μ combo	4.428	4.228	4.628	DQY < 0 at low	\$37,797
OS μ pembro	3.725	3.525	3.925	59,135–57,130	\$30,496

The ICER remained below \$60,000/QALY across all tested ranges.

8.2 Probabilistic Sensitivity Analysis

8.2.1 PSA Implementation Details

PSA with 1,000 Monte Carlo iterations (seed = 42). Survival parameters (μ , σ for 6 curves) drawn jointly from bivariate normal via Jacobian covariance ($r > 0.95$). Costs: gamma (CV=0.2). Utilities: beta (SE=0.03). Ordering constraints (RFS $\leq$ DMFS $\leq$ OS) verified every cycle; 828/1000 iterations valid.

Table S12: PSA Parameter Distributions and Sampled Values

Parameter	Distribution	Fitted mean	SE	Correlation	Sample
OS μ / σ – combo	Bivariate normal	8.396 / 3.096	0.793 / 0.522	0.996	Joint
OS μ / σ – pembro	Bivariate normal	4.841 / 1.131	0.291 / 0.310	0.955	Joint
r _{DM} μ / σ – combo	Bivariate normal	7.113 / 2.480	0.814 / 0.563	0.995	Joint
r _{DM} μ / σ – pembro	Bivariate normal	7.042 / 5.000	2.278 / 3.130	0.995	Joint
r _{LR} μ / σ – combo	Bivariate normal	8.874 / 5.000	2.039 / 1.928	0.997	Joint
r _{LR} μ / σ – pembro	Bivariate normal	5.913 / 2.674	0.757 / 0.879	0.984	Joint
Intismeran cost	Gamma (CV=0.2)	\$200,000	—	—	Independent
Pembrolizumab annual cost	Gamma (CV=0.2)	\$220,896	—	—	Independent
LR monthly cost	Gamma (CV=0.2)	\$3,000	—	—	Independent
DM monthly cost	Gamma (CV=0.2)	\$12,000	—	—	Independent
Utility RF / LR / DM	Beta (SE=0.03)	0.83 / 0.64 / 0.55	—	—	Independent

8.2.2 PSA Results

Mean Δ QALY = 4.04 (95% CI 2.29–6.71), mean Δ Cost = \$208,693, mean ICER = \$51,574 (median \$45,724). P(CE) = 56.2% at \$50K, 93.7% at \$100K, 98.2% at \$150K.

8.2.3 PSA Convergence

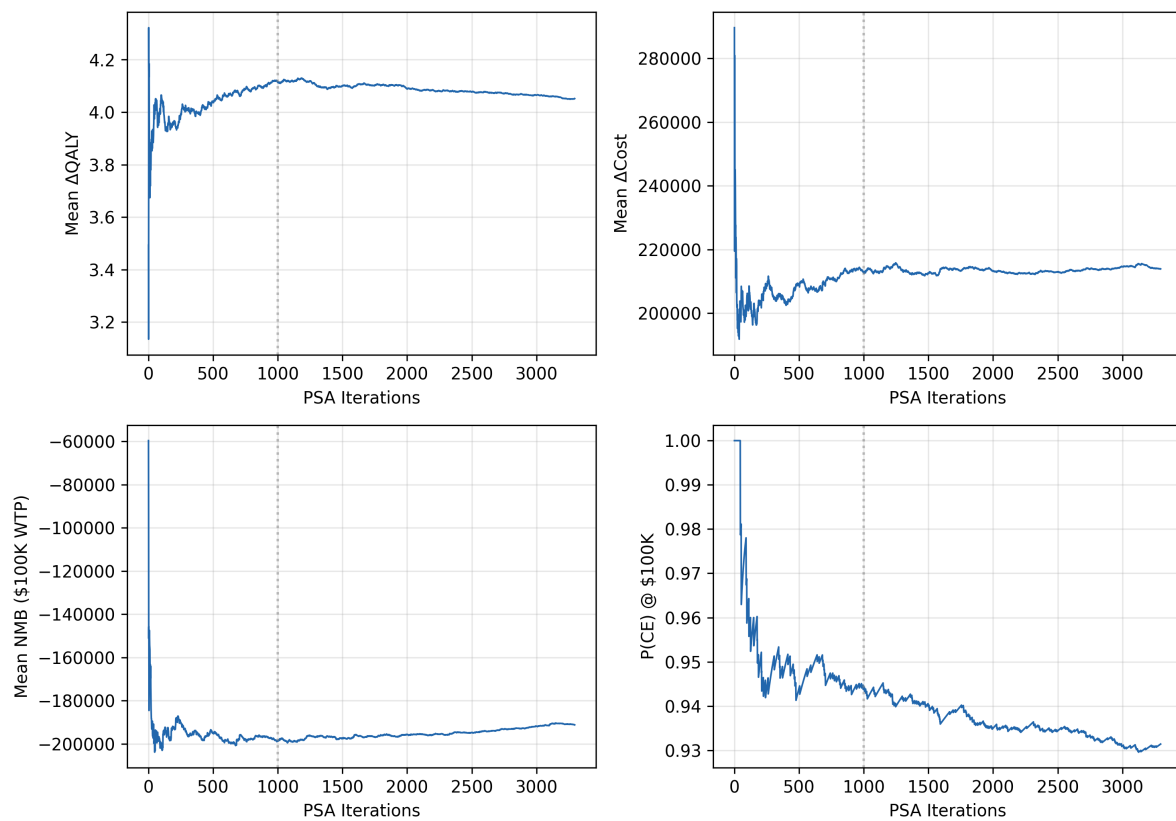


Figure S6: PSA convergence across iteration counts. Cumulative means of $\Delta QALY$, $\Delta Cost$, NMB, and P(CE). Vertical dotted line at 1,000 iterations. Estimates stable from ~ 500 iterations.

8.2.4 Cost-Effectiveness Plane

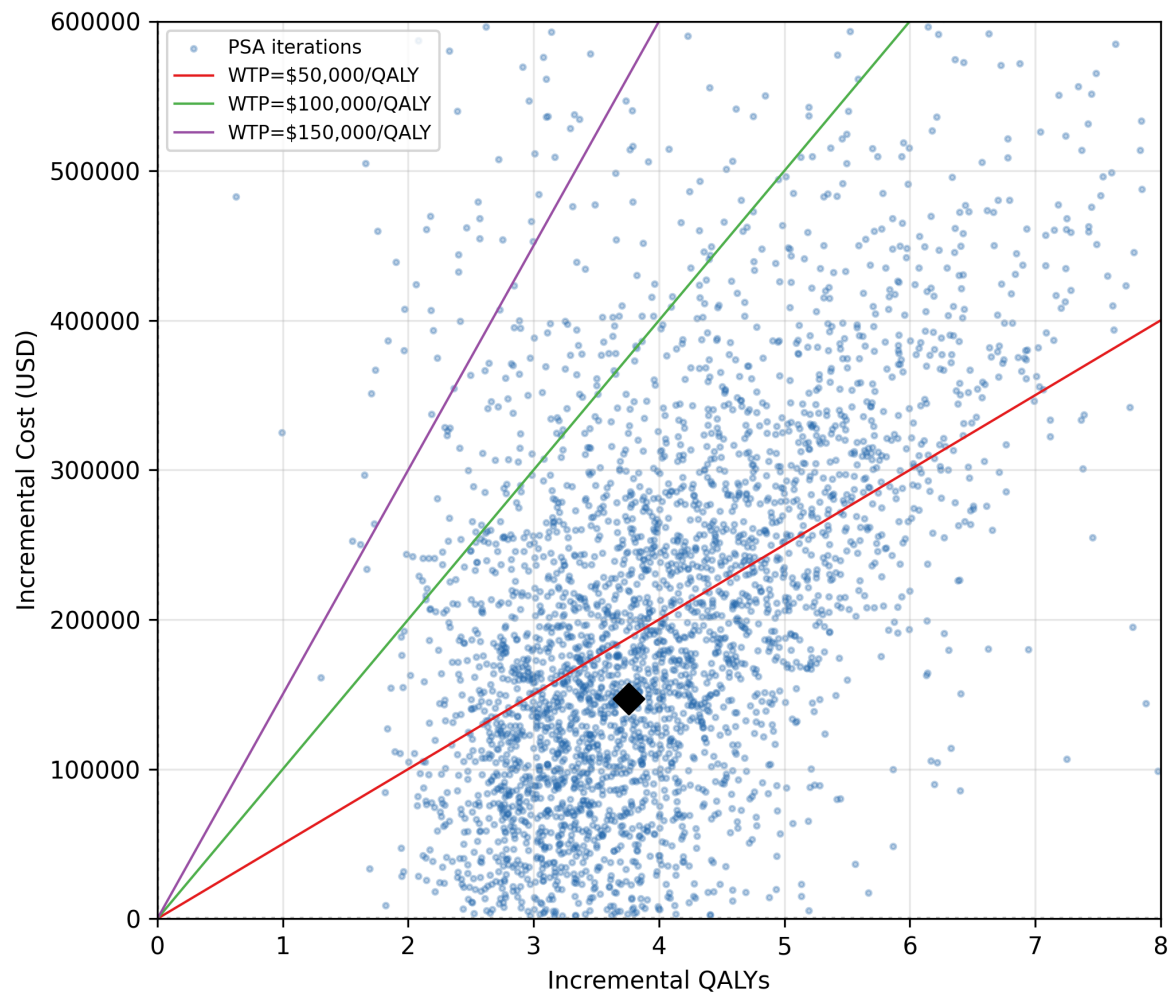


Figure S7: CE plane showing 1,000 PSA iterations (blue), WTP lines, and base case (black diamond).

8.3 Structural Scenario Analyses

Table S13: Structural Scenario Analysis Results

Scenario	ΔLY	$\Delta QALY$	$\Delta Cost$	ICER	NMB \$100K	NMB \$150K
Base case	4.42	3.76	\$146,961	\$39,105	\$228,851	\$416,7
Weibull OS	6.77	5.35	\$293,679	\$54,921	\$241,047	\$508,4
Unconstrained (no GP)	9.24	7.22	\$373,488	\$51,753	\$348,189	\$709,0
GP floor 48 mo	4.37	3.72	\$145,054	\$39,014	\$226,750	\$412,6
GP floor 60 mo (base)	4.42	3.76	\$146,961	\$39,105	\$228,851	\$416,7
GP floor 72 mo	4.50	3.81	\$149,673	\$39,251	\$231,648	\$422,3
10-year horizon	1.27	1.28	\$95,573	\$74,945	\$31,951	\$95,7
20-year horizon	3.32	2.91	\$115,739	\$39,734	\$175,548	\$321,1
40-year horizon (base)	4.42	3.76	\$146,961	\$39,105	\$228,851	\$416,7
Intismeran \$100K	4.42	3.76	\$46,961	\$12,496	\$328,851	\$516,7
Intismeran \$200K (base)	4.42	3.76	\$146,961	\$39,105	\$228,851	\$416,7
Intismeran \$300K	4.42	3.76	\$246,961	\$65,714	\$128,851	\$316,7
Intismeran \$500K	4.42	3.76	\$446,961	\$118,932	-\$71,149	\$116,7
Waning V1 (recurrence)	1.83	1.87	\$38,544	\$20,591	\$148,647	\$242,2
Waning V2 (hazard convergence)	2.66	2.47	\$74,620	\$30,176	\$172,659	\$296,2
No direct OS benefit	0.00	0.50	-\$14,562	Dominant	\$64,131	\$88,9

The base-case conclusion is robust across structural assumptions. The ICER alone is insufficient: unconstrained model has moderate ICER (\$51,753) but implausible \$373,488 cost.

9 Value-Based Pricing

9.1 Price Threshold Analysis

Table S14: Price Threshold Analysis

Intismeran Price	ICER	NMB @ \$100K	NMB @ \$150K
\$0	-\$14,113	\$428,851	\$616,757
\$50,000	-\$809	\$378,851	\$566,757
\$100,000	\$12,496	\$328,851	\$516,757
\$150,000	\$25,800	\$278,851	\$466,757
\$200,000 (base)	\$39,105	\$228,851	\$416,757
\$250,000	\$52,409	\$178,851	\$366,757
\$300,000	\$65,714	\$128,851	\$316,757
\$350,000	\$79,019	\$78,851	\$266,757
\$400,000	\$92,323	\$28,851	\$216,757
\$429,077	\$100,060	-\$226	\$187,680
\$450,000	\$105,628	-\$21,149	\$166,757
\$500,000	\$118,932	-\$71,149	\$116,757
\$550,000	\$132,237	-\$121,149	\$66,757
\$600,000	\$145,541	-\$171,149	\$16,757
\$616,455	\$149,920	-\$187,604	\$302
\$650,000	\$158,846	-\$221,149	-\$33,243
\$700,000	\$172,150	-\$271,149	-\$83,243
\$750,000	\$185,455	-\$321,149	-\$133,243
\$800,000	\$198,759	-\$371,149	-\$183,243

Max price @ \$100K/QALY: **\$429,077** ($2.1 \times$ base). Max price @ \$150K/QALY: **\$616,455** ($3.1 \times$ base).

9.2 ICER versus Acquisition Price

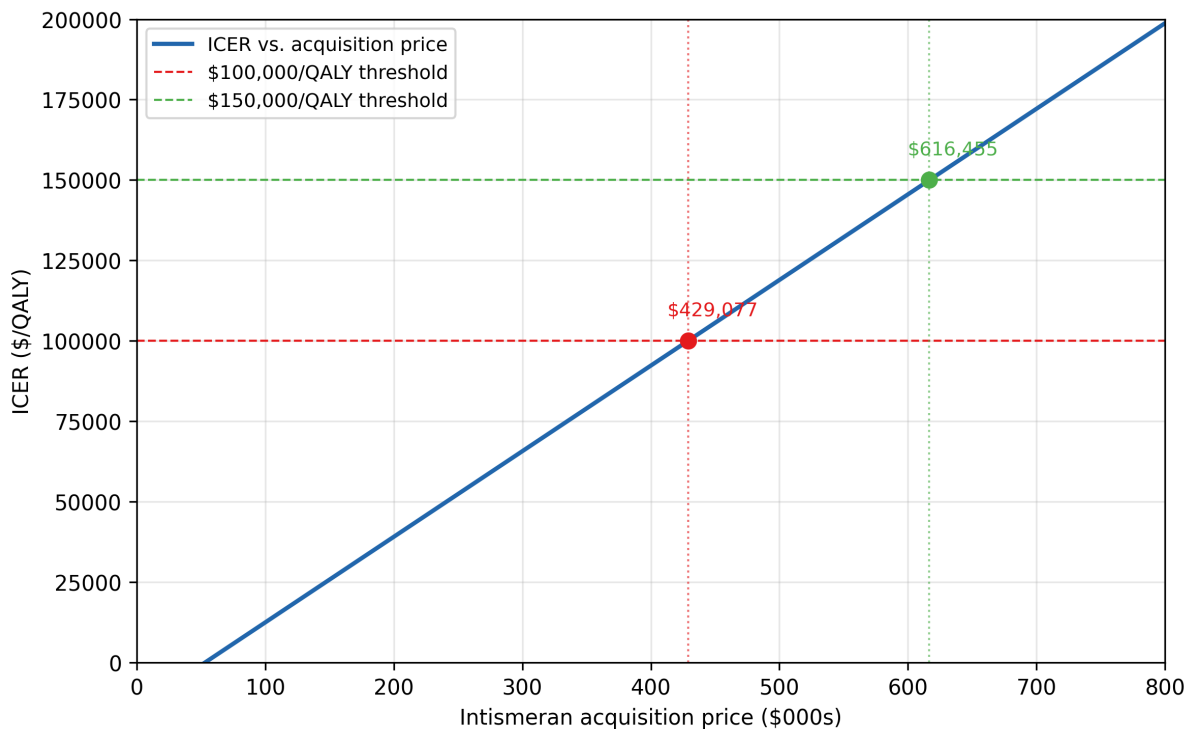


Figure S8: ICER as a function of intismeran acquisition price. Red/green dashed lines: \$100K/\$150K thresholds. Vertical dotted lines: threshold prices.

10 Reproducibility and Reporting

10.1 Technical Reproducibility

All analyses were performed using Python 3.12.7 (Apple Clang 15.0.0) on macOS 26.6.2 (Apple Silicon). The full code repository is available at <https://github.com/yichao2022/cea-intismeran> (tag v1.0-submission).

Key dependencies: NumPy 1.26.4, SciPy 1.14.1, Matplotlib 3.9.3, Pandas 2.2.3, LaTeX (TeX Live 2025, pdfLaTeX + BibTeX + natbib).

Model details:

- Cycle length: monthly (1/12 year); time horizon: 40 years (480 cycles)
- Half-cycle correction: not applied (monthly cycles are sufficiently fine-grained)
- Discounting: continuous-time exponential: $1/(1+r)^t$ where $r = 0.03$
- Random seed: 42 (PSA reproducibility)
- GP mortality source: US 2019 period life tables (NVSS, Arias et al. 2022)

Code structure:

- `model.py` — ModelParams, CEAModel (4-state PSM)
- `rerun_primary.py` — CEAModelV2 (hazard-based GP floor, waning, no-OS)
- `psa_full.py` — PSA with joint survival sampling

- `survival_model_selection.py` — 4,096-combination model selection
- `final_run.py` / `threshold_analysis.py` — DSA, scenarios, pricing
- `figures.py` / `gen_fig*.py` — Figure generation scripts
- `output/survival_combos_4096.csv` — Full model selection results

Data extraction: Survival probabilities extracted from Figure 1 of Khattak et al. (2026) using WebPlotDigitizer v4.7. The digitized values are in Table S2.

Reproducibility: Running `python3 rerun_primary.py` reproduces base-case results. `python3 psa_full.py` reproduces PSA. GitHub tag `v1.0-submission` freezes the submission code.

10.2 CHEERS 2022 Checklist

Table S15: CHEERS 2022 Checklist

No.	Item	Reported Location	
1	Title	Yes	Title page
2	Abstract	Yes	Abstract
3	Background	Yes	Introduction
4	Health economic analysis plan	Yes	Methods
5	Model	Yes	Methods, Appendix A
6	Population	Yes	Methods
7	Setting	Yes	Methods
8	Comparators	Yes	Methods
9	Perspective	Yes	Methods
10	Time horizon	Yes	Methods
11	Discount rate	Yes	Methods
12	Outcomes	Yes	Methods
13	Health states	Yes	Appendix A
14	Measurement of outcomes	Yes	Methods
15	Resource use	Yes	Methods
16	Currency	Yes	Methods
17	Model choice	Yes	Methods
18	Assumptions	Yes	Methods
19	Analytic methods	Yes	Methods
20	Study parameters	Yes	Methods, Appendix D
21	Incremental results	Yes	Results
22	Uncertainty	Yes	PSA, DSA
23	Heterogeneity	Yes	Subgroup
24	Discussion	Yes	Discussion
25	Limitations	Yes	Discussion
26	Funding	Yes	Declarations
27	Conflicts	Yes	Declarations
28	Reproducibility	Yes	Appendix

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